

Lab Exercise 6: Count Pairs with a Given Difference

Objective

To write a Java program that counts the number of pairs in an array whose absolute difference equals a given integer **k**.

Aim

To implement a Java function that efficiently counts the number of pairs (a, b) in an array such that

$$|a - b| = k$$

Theory

The **difference pair problem** is a common array-based question that tests understanding of loops, hashing, and efficient searching.

The brute-force method uses nested loops, leading to **O(n²)** time complexity.

However, using a **HashSet** or **HashMap** can reduce the complexity to **O(n)** by allowing constant-time lookups.

Algorithm / Procedure

1. Input:
 - An integer **n** representing the size of the array.
 - **n** space-separated integers (array elements).
 - An integer **k**, representing the difference.
 2. Initialize a **HashSet** to store array elements.
 3. For each element **x** in the array:
 - Check if (x + k) or (x - k) exists in the set.
 - Count such pairs.
 4. Print the total count of valid pairs.
-

Program Code: Problem1_CountPairs.java

```
import java.util.*;
```

```
public class Problem1_CountPairs {
```

```
// Function to count pairs with difference k
public static int countPairsWithDifference(int[] arr, int k) {
    // TODO: Implement your logic here
    return 0; // placeholder
}

public static void main(String[] args) {
    Scanner sc = new Scanner(System.in);

    System.out.print("Enter array size: ");
    int n = sc.nextInt();
    int[] arr = new int[n];

    System.out.println("Enter array elements:");
    for (int i = 0; i < n; i++) {
        arr[i] = sc.nextInt();
    }

    System.out.print("Enter difference k: ");
    int k = sc.nextInt();

    int result = countPairsWithDifference(arr, k);
    System.out.println("Number of pairs with difference " + k + ": " + result);

    sc.close();
}
}
```

Sample Input / Output

Input:

5

1 5 3 4 2

2

Output:

3

Explanation:

The valid pairs are (1,3), (5,3), and (4,2).

Evaluation Criteria (Total Marks: 10)

Test	Description	Marks
Program compiles and runs successfully	No syntax or runtime errors	2
Correct logic for counting pairs	Accurate result for all inputs	3
Proper use of HashSet/HashMap	Efficient implementation	2
Input/output handling	Matches expected format	2
Code readability and comments	Proper indentation and clarity	1
Total		10

Result

The program successfully counts all unique pairs in the array having the specified difference **k**.